

Deconvolving the age–metallicity distribution of APOGEE DR17 giants:

the current extreme-deconvolution model, its failure mode,
and a proposal for a calibrated model

Eos project working note

25 September 2026

Summary

- We fit the joint distribution of AstroNN age and APOGEE [Fe/H] for 164,215 giants with extreme deconvolution (XD; [Bovy et al., 2011](#)): a Gaussian mixture convolved with each star’s measurement covariance, with the number of components K chosen by five-fold held-out log density.
- The fitted *deconvolved* density is made of needle-like components: 17 of 20 components at $K = 20$ have intrinsic age widths below 0.45 Gyr, while the median quoted age error is 1.29 Gyr.
- Cause: the XD noise model, $\hat{a} = a + \epsilon$, is contradicted by the data. For high- α stars, the observed age scatter is *smaller* than the quoted total age error in every [Fe/H] bin (0.9–1.4 Gyr vs 1.6–3.2 Gyr). This is the signature of a shrinkage estimator, not additive noise. XD can satisfy it only by sending intrinsic age variances towards zero.
- Held-out likelihood cannot detect this. It scores the noise-convolved model, which is insensitive to intrinsic structure below the noise scale.
- Proposal: fit in $(\ln a, [\text{Fe}/\text{H}])$; replace the quoted AstroNN errors with a *calibrated linear-Gaussian measurement model* $\hat{\mathbf{y}} = \mathbf{R}\mathbf{v} + \mathbf{c} + \boldsymbol{\epsilon}$, derived from asteroseismic ages (APOKASC-3; [Pinsonneault et al., 2025](#)). XD handles this exactly through its projection matrices. Add bounded support, a covariance prior, paired cross-validation, and injection–recovery validation.

1 Data and sample

The sample is the base selection of the Eos figures repository (`eos_figures/data.py`): APOGEE DR17 `allStarLite` ([Abdurro’uf et al., 2022](#)) matched to the AstroNN DR17 value-added catalogue ([Leung & Bovy, 2019](#); [Mackereth et al., 2019](#)). It is restricted to red giants ($\log g < 3$) with velocity errors $< 50 \text{ km s}^{-1}$, abundance errors $< 0.2 \text{ dex}$, heliocentric distance $< 15 \text{ kpc}$, no Magellanic Cloud targets, and no stars near known satellites or in the Vasiliev globular-cluster member list. This leaves 210,392 stars.

The *good-age* subsample (mask `base_age`) further requires a finite, positive AstroNN age and $\sigma_{\text{model}}/a < 0.2$, where σ_{model} is `age_model_error`. It contains 164,215 stars. Figure 1 shows its density in the age–[Fe/H] plane.

The deconvolution uses two error columns:

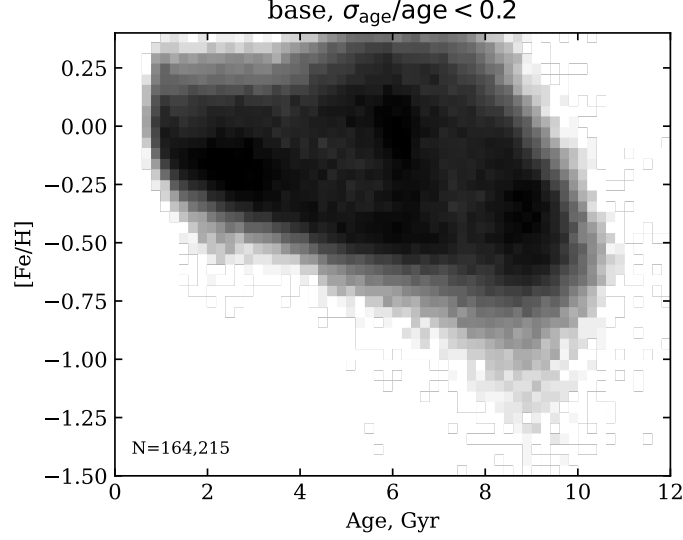


Figure 1: Observed density of the 164,215 good-age stars: $\log_{10}$ counts in $0.2 \text{ Gyr} \times 0.045 \text{ dex}$ bins, grey stretch between the 10th and 99th percentiles.

- **age_total_error** (σ_{tot}), AstroNN’s combined model and propagated-measurement uncertainty. Among good-age stars, $\sigma_{\text{tot}}/\sigma_{\text{model}}$ has 1st, 16th, 50th, 84th and 99th percentiles of 1.01, 1.12, 1.63, 2.45 and 4.58.
- **fe_h_err**, APOGEE’s $[\text{Fe}/\text{H}]$ error, with median 0.009 dex.

In practice the $[\text{Fe}/\text{H}]$ dimension is noiseless compared with the age dimension.

2 The current model

2.1 Extreme deconvolution

Let $\mathbf{v}_i = (a_i, [\text{Fe}/\text{H}]_i)$ be the true age and metallicity of star i , $\mathbf{w}_i$ the measured values, and $\mathbf{S}_i$ the measurement covariance. XD assumes

$$\mathbf{w}_i = \mathbf{v}_i + \boldsymbol{\epsilon}_i, \quad \boldsymbol{\epsilon}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{S}_i), \quad p(\mathbf{v}) = \sum_{k=1}^K \alpha_k \mathcal{N}(\mathbf{v} \mid \boldsymbol{\mu}_k, \mathbf{V}_k), \quad (1)$$

so that the likelihood of the data is

$$\ln \mathcal{L} = \sum_i \ln \sum_k \alpha_k \mathcal{N}(\mathbf{w}_i \mid \boldsymbol{\mu}_k, \mathbf{T}_{ik}), \quad \mathbf{T}_{ik} = \mathbf{V}_k + \mathbf{S}_i. \quad (2)$$

The EM iteration of [Bovy et al. \(2011\)](#) alternates two steps. The E step computes the responsibilities $q_{ik} \propto \alpha_k \mathcal{N}(\mathbf{w}_i \mid \boldsymbol{\mu}_k, \mathbf{T}_{ik})$ and the conditional moments

$$\mathbf{b}_{ik} = \boldsymbol{\mu}_k + \mathbf{V}_k \mathbf{T}_{ik}^{-1} (\mathbf{w}_i - \boldsymbol{\mu}_k), \quad \mathbf{B}_{ik} = \mathbf{V}_k - \mathbf{V}_k \mathbf{T}_{ik}^{-1} \mathbf{V}_k. \quad (3)$$

The M step updates

$$\alpha_k = \frac{1}{n} \sum_i q_{ik}, \quad \boldsymbol{\mu}_k = \frac{\sum_i q_{ik} \mathbf{b}_{ik}}{\sum_i q_{ik}}, \quad \mathbf{V}_k = \frac{\sum_i q_{ik} [(\boldsymbol{\mu}_k - \mathbf{b}_{ik})(\boldsymbol{\mu}_k - \mathbf{b}_{ik})^\top + \mathbf{B}_{ik}]}{\sum_i q_{ik}}. \quad (4)$$

2.2 Implementation

The code, `eos_figures/xd.py`, is a port of the numpy XD in the `qso_p_color` project, restricted to fully observed data:

- **2D fast path.** A closed-form element-wise 2×2 E/M step matches the generic batched algebra to 2×10^{-13} , and `log_gauss` matches `scipy` to 2×10^{-14} . At $K = 8$, $n = 164,215$ it reduces the time per iteration from 0.64s to 0.076s.
- **Numerical care.** Second moments are accumulated about the previous mean, and the reported likelihood is recomputed for the returned mixture.
- **Noise model.** $\mathbf{S}_i = \text{diag}(\sigma_{\text{tot},i}^2, \sigma_{[\text{Fe}/\text{H}],i}^2)$, with no age-[Fe/H] error covariance.
- **Convergence.** At $K = 8$ with no tolerance, the mean log-likelihood per star still rose by 2.2×10^{-3} after iteration 2000, alternating plateaus and jumps. A single-step tolerance therefore stops too early. We stop when the mean per-star gain, averaged over the last 50 iterations, falls below $3 \times 10^{-7} |\ln \mathcal{L}/n|$ (about 0.1 in total $\ln \mathcal{L}$ per iteration), with at most 3000 iterations.
- **Initialisation.** At $K = 8$ we tested three seeds each. *k*-means++ starts converge in about 730 iterations but reach $\ln \mathcal{L}/n = -2.2475$. Random-row starts (the `qso_p_color` default) take 1200–1500 iterations but reach -2.2447 , better by 2.8×10^{-3} per star (≈ 460 in total). Random starts are used.
- **Tests.** Nine tests: `tests/test_xd.py` (recovering intrinsic widths; two-component recovery; heteroscedastic anisotropic noise; monotone likelihood; CV complexity choice; BIC penalty) and `tests/test_xd_age_feh.py` (grid counts integrate to N ; the error-grouping approximation matches the exact per-star sum to $< 1\%$ of the peak; convolved counts match a 2×10^5 -star Monte Carlo, with pull mean < 0.3 and pull width in 0.7–1.3).

2.3 Model selection

Following `qso_p_color`, K is chosen to maximise the five-fold held-out log density of the *observed* data under the convolved model, eq. (2). Each (K , fold) pair uses two random restarts, and the restart with the higher *training* likelihood is kept, so model choice never sees the validation data.

BIC and AIC are computed from full-sample fits for comparison. They are not used for the decision, because mixtures are singular models and the regularity conditions behind BIC fail.

The runner, `scripts/fit_xd_age_feh.py`, is parallel and resumable, writing one JSON file per job.

Table 1 gives the held-out scores for the values of K completed so far. They rise steeply up to $K \approx 12$ and flatten beyond $K \approx 16$. The quoted standard error (1.76×10^{-3}) is that of each K 's mean on its own; it is dominated by star-to-star scatter shared by all K . Differences between K need *paired* per-star comparisons, which the proposed model stores (Sec. 4.6).

2.4 Evaluating the model on the plotting grid

The *deconvolved* model gives expected counts $n \int_{\text{bin}} p(\mathbf{v}) d\mathbf{v}$, integrated by the midpoint rule on 5×5 sub-cells per bin. The *convolved* model gives $\sum_i \int_{\text{bin}} \sum_k \alpha_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \mathbf{V}_k + \mathbf{S}_i) d\mathbf{x}$. For speed, stars are grouped on a 40×40 grid in $(\ln \sigma_a, \ln \sigma_{[\text{Fe}/\text{H}]})$, and each group uses its mean variances; the tests above bound the resulting error. Both are implemented in `eos_figures/xd_age_feh.py`.

2.5 Result: a needle-like deconvolved density

Figure 2 shows the deconvolved density of the best available $K = 20$ model. This is fold model `K20_f4_r0`, trained on 80% of the stars; it scores highest of the ten $K = 20$ fold models when evaluated on all stars ($\ln \mathcal{L}/n = -2.23991$). Table 2 lists its components:

Table 1: Five-fold held-out mean log density per star (natural log; units of $\ln[\text{Gyr}^{-1}\text{dex}^{-1}]$) for the current model. Δ is relative to the best completed K . All folds converged. $K \leq 7$ was still running when this note was written.

K	CV	Δ	K	CV	Δ
8	-2.24596	-0.00557	15	-2.24102	-0.00063
9	-2.24347	-0.00308	16	-2.24059	-0.00020
10	-2.24213	-0.00174	17	-2.24051	-0.00012
11	-2.24160	-0.00121	18	-2.24069	-0.00030
12	-2.24123	-0.00084	19	-2.24066	-0.00027
13	-2.24110	-0.00071	20	-2.24039	0
14	-2.24089	-0.00050			

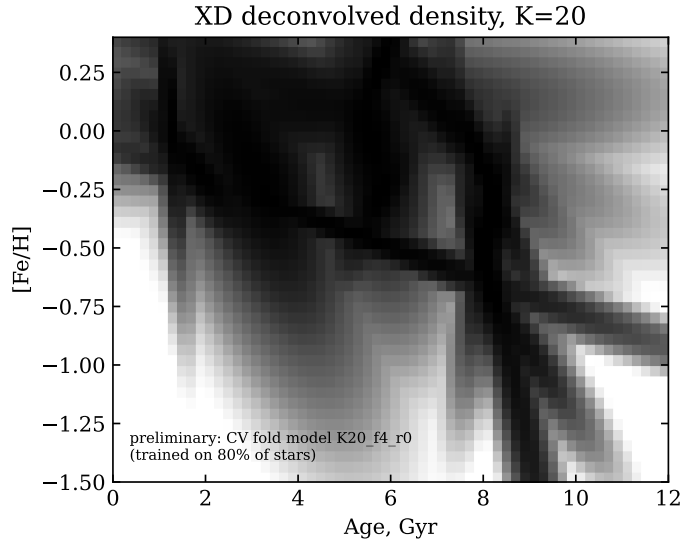


Figure 2: Deconvolved density of the current model (preliminary $K = 20$ fold model), shown as expected counts per bin on the grid of Fig. 1.

- 17 of the 20 components have $\sigma_a < 0.45$ Gyr, and several are almost degenerate lines ($|\rho|$ up to 0.99).
- The three heaviest components near $a \approx 8.1$ Gyr carry 28% of the weight, with $\sigma_a = 0.11$ –0.24 Gyr.

These widths are far below anything the data could resolve, given $\sigma_{\text{tot}} \sim 1$ –3 Gyr.

3 Why a better model is needed

3.1 What the intrinsic density should look like

Established Milky Way structure suggests three components with distinct shapes in the age–[Fe/H] plane:

1. **Low- α (thin) disc:** ages ~ 0 –8 Gyr and $-0.7 \lesssim [\text{Fe}/\text{H}] \lesssim +0.4$. The mean age–metallicity relation is nearly flat, with a large spread in [Fe/H] at fixed age (~ 0.2 dex, in part from radial migration). The density should be broad and smooth on scales $\gtrsim 1$ Gyr, apart from possible modest star-formation bursts.

Table 2: Components of the $K = 20$ model of Fig. 2, sorted by weight. σ_a and σ_{Fe} are intrinsic widths (Gyr, dex); ρ is the age–[Fe/H] correlation.

α_k	μ_a [Gyr]	μ_{Fe}	σ_a [Gyr]	σ_{Fe}	ρ
0.142	2.92	−0.12	0.392	0.160	−0.67
0.114	8.17	−0.40	0.110	0.107	+0.74
0.095	5.20	−0.44	1.221	0.088	−0.99
0.092	8.09	−0.60	0.138	0.109	−0.83
0.079	8.00	−0.16	0.238	0.116	−0.95
0.065	5.53	+0.00	0.254	0.094	−0.31
0.060	5.77	+0.26	0.190	0.087	+0.82
0.056	1.84	−0.17	0.337	0.083	−0.87
0.048	5.77	−0.28	0.227	0.079	+0.75
0.035	3.43	−0.24	0.244	0.088	−0.06
0.031	1.27	+0.05	0.060	0.116	−0.68
0.031	7.29	+0.03	0.399	0.080	−0.88
0.027	6.61	+0.18	0.442	0.078	−0.84
0.025	4.16	+0.08	1.039	0.052	+0.15
0.025	5.65	−0.11	0.224	0.118	−0.34
0.020	5.59	−0.10	0.267	0.120	−0.23
0.018	8.33	−0.71	0.303	0.292	−0.97
0.018	5.65	−0.14	0.241	0.108	−0.12
0.014	4.02	+0.17	0.949	0.081	−0.84
0.004	8.67	−0.86	0.273	0.077	−0.93

2. **High- α (thick) disc:** ages ~ 8 – 12 Gyr, following a tight, tilted age–metallicity sequence in which [Fe/H] rises from ~ -1 to ~ 0 as age decreases (Haywood et al., 2013). A narrow, negatively correlated ridge is physical here.
3. **Old metal-poor populations:** the accreted GS/E debris (Belokurov et al., 2018; Helmi et al., 2018) and the in-situ *Aurora* population (Belokurov & Kravtsov, 2022), at ages $\gtrsim 10$ Gyr and [Fe/H] $\lesssim -0.7$.

The support is bounded: $0 < a \lesssim 13.8$ Gyr. The density also includes the selection function: APOGEE targeting, and the number of giants per unit stellar mass formed, which depends on age. It is not the star-formation history.

3.2 The additive-noise model is contradicted by the data

If $\hat{a} = a + \epsilon$ with ϵ independent of a and $\text{Var}(\epsilon) = \sigma^2$, then within any subsample

$$\text{Var}(\hat{a}) = \text{Var}(a) + \langle \sigma^2 \rangle \geq \langle \sigma^2 \rangle. \quad (5)$$

Figure 3 tests this in 0.1-dex [Fe/H] bins; Table 3 gives 0.2-dex values:

- For high- α stars the inequality fails with σ_{tot} in *every* bin: $\text{std}(\hat{a}) = 0.9$ – 1.4 Gyr against rms $\sigma_{\text{tot}} = 1.6$ – 3.2 Gyr.
- For all good-age stars it holds only for $-0.4 < [\text{Fe}/\text{H}] < +0.2$.
- Even the smaller σ_{model} violates it for $[\text{Fe}/\text{H}] < -0.8$ (all good-age stars) and $[\text{Fe}/\text{H}] < -0.6$ (high- α).

Under eq. (1), the maximum-likelihood response to $\langle \sigma^2 \rangle > \text{Var}(\hat{a})$ is $\mathbf{V}_k^{aa} \rightarrow 0$: exactly the needles of Fig. 2.

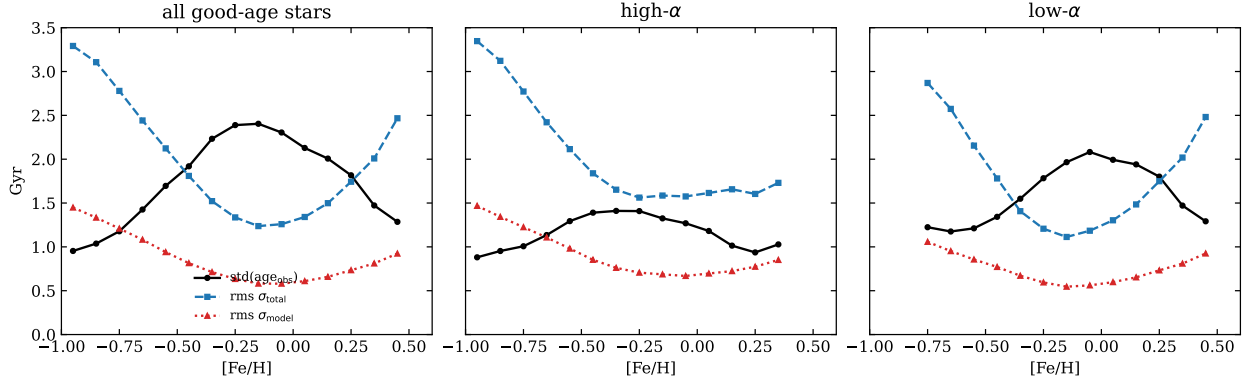


Figure 3: Observed age scatter (black) compared with the rms quoted total (blue) and model (red) AstroNN age errors, in 0.1-dex $[\text{Fe}/\text{H}]$ bins. Additive noise requires the black curve to lie above the error curves.

Table 3: Additive-noise test, eq. (5), in 0.2-dex bins: $\text{Var}(\hat{a}) - \langle \sigma^2 \rangle$ in Gyr^2 . Negative entries are inconsistent with additive noise.

$[\text{Fe}/\text{H}]$ bin	all good-age		high- α	
	σ_{tot}	σ_{model}	σ_{tot}	σ_{model}
−1.0 to −0.8	−8.99	−0.85	−9.29	−1.03
−0.8 to −0.6	−4.66	+0.59	−5.26	−0.12
−0.6 to −0.4	−0.31	+2.71	−2.06	+0.99
−0.4 to −0.2	+3.45	+5.04	−0.61	+1.45
−0.2 to 0.0	+4.00	+5.22	−0.80	+1.24
0.0 to +0.2	+2.35	+3.93	−1.38	+0.78
+0.2 to +0.4	−0.36	+2.41	−1.73	+0.33

3.3 Interpretation: AstroNN ages behave like shrinkage estimates

A regression network trained on asteroseismic ages (APOKASC-2; Pinsonneault et al., 2018) returns something closer to a posterior mean, $\hat{a} \approx \text{E}[a \mid \text{spectrum}]$, than to a noisy copy of a . By the law of total variance,

$$\text{Var}(\hat{a}) = \text{Var}(a) - \text{E}[\text{Var}(a \mid \text{spectrum})], \quad (6)$$

so the estimates are *under*-dispersed, and the quoted uncertainty measures posterior width, not scatter about the truth.

For a Gaussian prior (the training-set distribution) $\mathcal{N}(m_0, P)$ and a Gaussian likelihood of width s , the posterior mean is $\hat{a} = (1 - \lambda)a + \lambda m_0 + \eta$ with $\lambda = s^2/(s^2 + P)$. This is a linear map with slope $\beta = 1 - \lambda < 1$, plus noise.

Two further observations point the same way:

- **A ceiling.** Only 0.65% of good-age stars have $\hat{a} > 10$ Gyr and 0.02% have $\hat{a} > 11$ Gyr, with a maximum of 12.25 Gyr. This conflicts with the expected 10–13 Gyr high- α and halo populations (Sec. 3.1).
- **The largest errors fall where the training data are sparse.** The median σ_{tot} exceeds 2–3 Gyr at $[\text{Fe}/\text{H}] < -0.6$ and 7–11 Gyr (Fig. 4).

3.4 Further limitations of the current set-up

- **Linear age.** Errors are close to fractional (median $\sigma_{\text{tot}}/\hat{a} = 0.31, 0.28, 0.27, 0.26, 0.19, 0.18$ for 0–2, 2–4, 4–6, 6–8, 8–10 and 10–13 Gyr), so they are closer to Gaussian in $\ln a$. Gaussian components in linear a also put probability at $a < 0$ and above the age of the Universe.

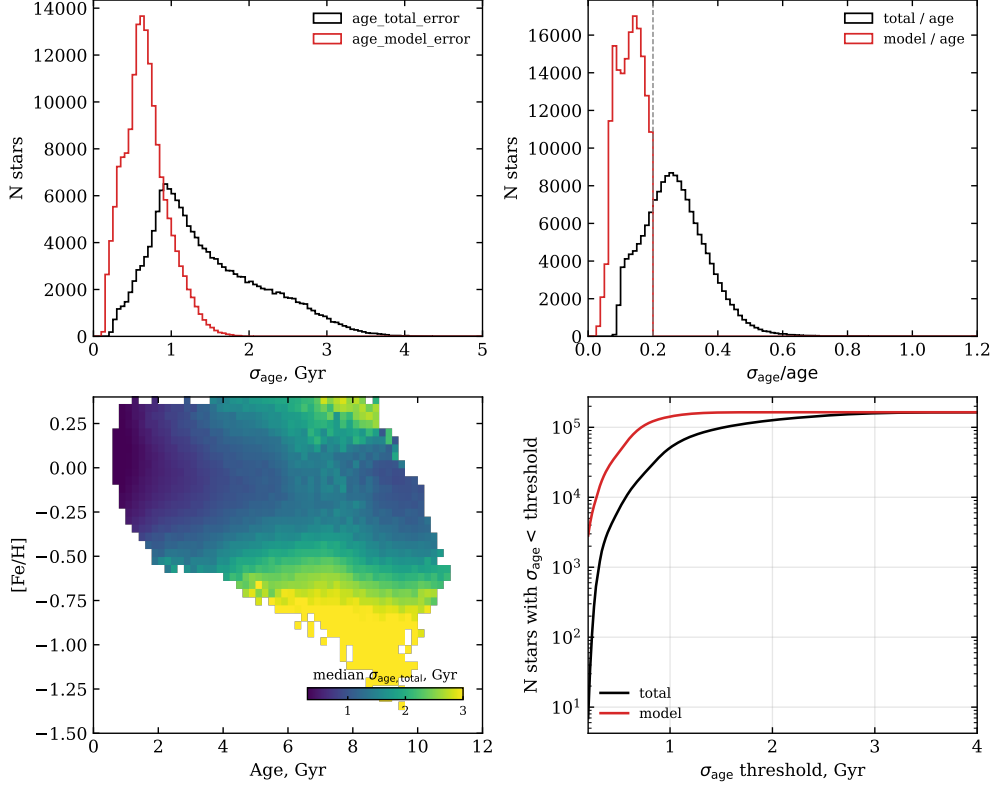


Figure 4: AstroNN age errors of the good-age sample: *top left*, distributions of σ_{tot} and σ_{model} ; *top right*, fractional errors, with the dashed line marking the $\sigma_{\text{model}}/a < 0.2$ selection; *bottom left*, median σ_{tot} across the age–[Fe/H] plane; *bottom right*, the number of stars below an error threshold.

- **Weak identifiability.** Held-out likelihood evaluates $\mathbf{V}_k + \mathbf{S}_i$. When $\mathbf{S}_i \gg \mathbf{V}_k$, very different intrinsic densities give almost the same score, so CV chooses K but cannot regularise intrinsic widths.
- **Selection on error.** Cutting on σ does not help. Stars with $\sigma_{\text{tot}} < 1$ Gyr include none with $[\text{Fe}/\text{H}] < -0.5$, and $\sigma_{\text{tot}} < 2.5$ Gyr keeps only 61% of $[\text{Fe}/\text{H}] < -0.5$ stars against 89% overall. The cut is a value-dependent selection that XD does not model.
- **Unpaired standard errors** in Table 1 overstate the uncertainty on differences between K values.

4 Proposed model

4.1 Variables

Fit $\mathbf{v} = (\ell, [\text{Fe}/\text{H}])$ with $\ell = \ln(a/\text{Gyr})$. The data are $\hat{\mathbf{y}}_i = (\ln \hat{a}_i, \widehat{[\text{Fe}/\text{H}]_i})$.

4.2 Calibrated measurement model

Replace eq. (1) by a linear-Gaussian measurement model,

$$\hat{\mathbf{y}}_i = \mathbf{R}_i \mathbf{v}_i + \mathbf{c}_i + \boldsymbol{\epsilon}_i, \quad \mathbf{R}_i = \begin{pmatrix} \beta(\mathbf{z}_i) & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{c}_i = \begin{pmatrix} \gamma(\mathbf{z}_i) \\ 0 \end{pmatrix}, \quad \boldsymbol{\epsilon}_i \sim \mathcal{N}(\mathbf{0}, \text{diag}[\tau^2(\mathbf{z}_i), \sigma_{[\text{Fe}/\text{H}],i}^2]). \quad (7)$$

Here $\mathbf{z}_i$ are observables on which the calibration may depend: $[\text{Fe}/\text{H}]$, $[\alpha/\text{Fe}]$, $\log g$, S/N, or the quoted σ_{model} .

The functions β (slope; $\beta < 1$ is shrinkage), γ (offset) and τ (true residual scatter) are *fitted* on stars with asteroseismic ages. We would use the APOKASC-3 catalogue (Pinsonneault et al., 2025), restricted to stars not used to train AstroNN or held out from its training. The calibration regression is itself an errors-in-variables fit, since asteroseismic ages have their own uncertainties. It is low-dimensional and can be done by maximum likelihood, e.g. piecewise-linear β, γ, τ in $[\text{Fe}/\text{H}]$ and $[\alpha/\text{Fe}]$.

Checks on the calibration.

- The additive test of Sec. 3.2 generalises to $\text{Var}(\hat{\ell}) = \beta^2 \text{Var}(\ell) + \langle \tau^2 \rangle$, which must hold in every bin.
- The calibration sample’s coverage in $[\text{Fe}/\text{H}]$ and age must be reported. Outside it, the model extrapolates, and the fitted density must be flagged there.

4.3 XD with projection

Equation (7) is exactly the form handled by XD through per-object projection matrices (Bovy et al., 2011, their Sec. 2):

$$\mathbf{T}_{ik} = \mathbf{R}_i \mathbf{V}_k \mathbf{R}_i^\top + \mathbf{S}_i, \quad (8)$$

$$\mathbf{b}_{ik} = \boldsymbol{\mu}_k + \mathbf{V}_k \mathbf{R}_i^\top \mathbf{T}_{ik}^{-1} (\hat{\mathbf{y}}_i - \mathbf{R}_i \boldsymbol{\mu}_k - \mathbf{c}_i), \quad (9)$$

$$\mathbf{B}_{ik} = \mathbf{V}_k - \mathbf{V}_k \mathbf{R}_i^\top \mathbf{T}_{ik}^{-1} \mathbf{R}_i \mathbf{V}_k, \quad (10)$$

with the M step unchanged. With $\beta < 1$, the intrinsic spread needed to explain a given observed spread *increases*, removing the pressure towards $\mathbf{V}_k^{\ell\ell} \rightarrow 0$.

The change to `eos_figures/xd.py` is small: $\mathbf{R}_i$ is diagonal and 2×2 , so the element-wise fast path extends directly.

4.4 Bounded support

Truncate the mixture at $\ell_{\max} = \ln 13.8$. The likelihood of star i becomes $\int_{\ell < \ell_{\max}} \mathcal{N}(\hat{\mathbf{y}}_i | \mathbf{R}_i \mathbf{v} + \mathbf{c}_i, \mathbf{S}_i) p(\mathbf{v}) d\mathbf{v} / Z$, where $Z = \sum_k \alpha_k \Phi_k$ and Φ_k is the mass of component k below $\ell_{\max}$. For Gaussian components this needs only univariate normal CDFs.

As a cheaper first step: fit without truncation, report the leaked mass $1 - Z$, and implement truncation only if the leakage exceeds 1%.

4.5 Covariance prior

Use MAP-EM with an inverse-Wishart prior on each $\mathbf{V}_k$, so that the M step becomes $\mathbf{V}_k = (\sum_i q_{ik}[\dots] + \boldsymbol{\Psi}) / (\sum_i q_{ik} + \nu + 3)$. Bovy et al.’s regularisation parameter w corresponds to $\boldsymbol{\Psi} = w\mathbf{I}$.

The floor should be tied to what the calibrated data can resolve: $\Psi^{\ell\ell} \sim (0.05)^2$ (a 5% age width) and $\Psi^{\text{FeFe}} \sim (0.02 \text{ dex})^2$. The results must be shown to be stable when these values are halved and doubled.

4.6 Model selection and posterior-predictive checks

- **Choosing K .** Five-fold held-out log density as now, but storing per-star held-out $\ln p$ so that differences between K values carry *paired* standard errors. Choose the smallest K within one paired standard error of the maximum.
- **Checks.**
 1. data – convolved-model residual maps, with the χ^2 per bin;
 2. the calibrated variance test above, applied to model draws;
 3. marginal age distributions for the high- α , low- α and metal-poor subsets, data against model.

4.7 Validation by injection–recovery

Build a mock intrinsic density with the three components of Sec. 3.1:

- a broad thin disc;
- a tilted, narrow thick-disc sequence;
- an old metal-poor tail up to 13 Gyr.

Draw 164,215 stars, map them through eq. (7) with the calibrated β, γ, τ , fit, and compare the recovered deconvolved density with the truth. Metrics: the mean and width of ℓ at fixed $[\text{Fe}/\text{H}]$, component weights, and the Kullback–Leibler divergence on the grid.

Run the same mock through the *current* model (additive noise with σ_{tot}) to show that it produces needle artefacts. This makes the failure mode demonstrable rather than inferred.

4.8 Cost

The current runner does 200 CV fits at $K \leq 20$ in ~ 2 hours on 12 cores when the machine is not oversubscribed. With a calibrated model, K is expected to be well below 20. With early stopping on the held-out score as K increases, CV should need well under an hour.

5 Next steps

1. Stop the current CV run, which uses a contradicted noise model, or keep it only as the “current model” baseline for the injection–recovery comparison.
2. Cross-match the sample with APOKASC-3; fit β, γ, τ and repeat the variance test with the calibrated model.
3. Implement projection XD in $(\ln a, [\text{Fe}/\text{H}])$ with an inverse-Wishart prior and paired CV; run injection–recovery; then refit the data and make the four-panel figure (data, deconvolved, convolved, residual).

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A Reproducing the numbers

```
source ~/Work/venvs/.venv/bin/activate
python -m pytest tests -q                                # 9 tests
python scripts/fit_xd_age_feh.py init-test               # initialisation test (K=8)
python scripts/fit_xd_age_feh.py cv --kmax 20 --init random
python scripts/fit_xd_age_feh.py full --kmax 20 --init random
python scripts/fit_xd_age_feh.py select                  # CV/BIC/AIC; model
python scripts/plot_eos_age_feh_density.py
python scripts/plot_eos_age_err_dist.py
python scripts/plot_eos_age_err_consistency.py
```